[Establishing transcriptional silencing of the X chromosome during early embryogenesis].

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Early development of female mammals is accompanied by transcriptional inactivation of one of their two X chromosomes. This process, known as X-chromosome inactivation, relies on monoallelic activation of the Xist gene. Xist produces a non-coding RNA that can coat the chromosome from which it is transcribed in cis and trigger its silencing. How Xist expression is controlled and how it initiates transcriptional repression are central questions for our understanding of how this chromosome-wide monoallelic program is expressed. Several trans-acting factors have been identified as regulators of Xist expression. Interestingly, some Xist activators are encoded by the X chromosome itself, thereby efficiently promoting Xist expression in females (XX) but not in males (XY). Female cells also display transient physical pairing between their two X chromosomes at the level of their Xics (X inactivation centers) during the time window when X inactivation is initiated. It has been proposed that these pairing events may play a role in Xist activation and its monoallelic regulation. Xist RNA accumulates over the X chromosome from which it is expressed and rapidly triggers the exclusion of the transcription machinery. Genic sequences are initially located outside of this Xist RNA coated domain but as they become progressively silenced they are relocated into this silent nuclear compartment created by Xist. However genes are not all silenced with the same kinetics. Furthermore, some genes can escape X inactivation and remain located outside the Xist-coated compartment. Recent findings have revealed that young, active LINE-1 retrotransposons are expressed from the inactive X chromosome and may facilitate X inactivation, particularly in regions of the X that would otherwise be prone to escape.

Crucial role of antisense transcription across the Xist promoter in Tsix-mediated Xist chromatin modification.

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Expression of Xist, which triggers X inactivation, is negatively regulated in cis by an antisense gene, Tsix, transcribed along the entire Xist gene. We recently demonstrated that Tsix silences Xist through modification of the chromatin structure in the Xist promoter region. This finding prompted us to investigate the role of antisense transcription across the Xist promoter in Tsix-mediated silencing. Here, we prematurely terminated Tsix transcription before the Xist promoter and addressed its effect on Xist silencing in mouse embryos. We found that although 93% of the region encoding Tsix was transcribed, truncation of Tsix abolished the antisense regulation of Xist. This resulted in a failure to establish the repressive chromatin configuration at the Xist promoter on the mutated X, including DNA methylation and repressive histone modifications, especially in extraembryonic tissues. These results suggest a crucial role for antisense transcription across the Xist promoter in Xist silencing.

AFF3-DNA methylation interplay in maintaining the mono-allelic expression pattern of XIST in terminally differentiated cells.

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X chromosome inactivation and genomic imprinting are two classic epigenetic regulatory processes that

cause mono-allelic gene expression. In female mammals, mono-allelic expression of the long non-coding RNA gene X-inactive specific transcript (XIST) is essential for initiation of X chromosome inactivation upon differentiation. We have previously demonstrated that the central factor of super elongation complex-like 3 (SEC-L3), AFF3, is enriched at gamete differentially methylated regions (DMRs) of the imprinted loci and regulates the imprinted gene expression. Here, we found that AFF3 can also bind to the DMR downstream of the XIST promoter. Knockdown of AFF3 leads to de-repression of the inactive allele of XIST in terminally differentiated cells. In addition, the binding of AFF3 to the XIST DMR relies on DNA methylation and also regulates DNA methylation level at DMR region. However, the KAP1-H3K9 methylation machineries, which regulate the imprinted loci, might not play major roles in maintaining the mono-allelic expression pattern of XIST in these cells. Thus, our results suggest that the differential mechanisms involved in the XIST DMR and gDMR regulation, which both require AFF3 and DNA methylation.

Mbd2 contributes to DNA methylation-directed repression of the Xist gene.

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Transcription of the Xist gene triggers X chromosome inactivation in cis and is therefore silenced on the X chromosome that remains active. DNA methylation contributes to this silencing, but the mechanism is unknown. As methylated DNA binding proteins (MBPs) are potential mediators of gene silencing by DNA methylation, we asked whether MBP-deficient cell lines could maintain Xist repression. The absence of Mbd2 caused significant low-level reactivation of Xist, but silencing was restored by exogenous Mbd2. In contrast, deficiencies of Mbd1, MeCP2, and Kaiso had no detectable effect, indicating that MBPs are not functionally redundant at this locus. Xist repression in Mbd2-null cells was hypersensitive to the histone deacetylase inhibitor trichostatin A and to depletion of the DNA methyltransferase Dnmt1. These synergies implicate Mbd2 as a mediator of the DNA methylation signal at this locus. The presence of redundant mechanisms to enforce repression at Xist and other loci is compatible with the hypothesis that "stacking" of imperfect repressive tendencies may be an evolutionary strategy to ensure leakproof gene silencing.

Silencing by imprinted noncoding RNAs: is transcription the answer?

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Non-coding RNAs (ncRNAs) with gene regulatory functions are starting to be seen as a common feature of mammalian gene regulation with the discovery that most of the transcriptome is ncRNA. The prototype has long been the Xist ncRNA, which induces X-chromosome inactivation in female cells. However, a new paradigm is emerging--the silencing of imprinted gene clusters by long ncRNAs. Here, we review models by which imprinted ncRNAs could function. We argue that an Xist-like model is only one of many possible solutions and that imprinted ncRNAs could provide the better model for understanding the function of the new class of ncRNAs associated with non-imprinted mammalian genes.

GATA transcription factors drive initial Xist upregulation after fertilization through direct activation of long-range enhancers.

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X-chromosome inactivation (XCI) balances gene expression between the sexes in female mammals. Shortly after fertilization, upregulation of Xist RNA from one X chromosome initiates XCI, leading to chromosome-wide gene silencing. XCI is maintained in all cell types, except the germ line and the pluripotent state where XCI is reversed. The mechanisms triggering Xist upregulation have remained elusive. Here we identify GATA transcription factors as potent activators of Xist. Through a pooled CRISPR activation screen in murine embryonic stem cells, we demonstrate that GATA1, as well as other GATA transcription factors can drive ectopic Xist expression. Moreover, we describe GATA-responsive regulatory elements in the Xist locus bound by different GATA factors. Finally, we show that GATA factors are essential for XCI induction in mouse preimplantation embryos. Deletion of GATA1/4/6 or GATA-responsive Xist enhancers in mouse zygotes effectively prevents Xist upregulation. We propose that the activity or complete absence of various GATA family members controls initial Xist upregulation, XCI maintenance in extra-embryonic lineages and XCI reversal in the epiblast.

Maintenance of X chromosome inactivation after T cell activation requires NF-κB signaling.

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X chromosome inactivation (XCI) balances X-linked gene dosage between sexes. Unstimulated T cells lack cytological enrichment of X-inactive specific transcript (Xist) RNA and heterochromatic modifications on the inactive X chromosome (Xi), which are involved in maintenance of XCI, and these modifications return to the Xi after stimulation. Here, we examined allele-specific gene expression and epigenomic profiles of the Xi in T cells. We found that the Xi in unstimulated T cells is largely dosage compensated and enriched with the repressive H3K27me3 modification but not the H2AK119-ubiquitin (Ub) mark. Upon T cell stimulation mediated by both CD3 and CD28, the Xi accumulated H2AK119-Ub at gene regions of previous H3K27me3 enrichment. T cell receptor (TCR) engagement, specifically NF-κB signaling downstream of the TCR, was required for Xist RNA localization to the Xi. Disruption of NF-κB signaling in mouse and human T cells using genetic deletion, chemical inhibitors, and patients with immunodeficiencies prevented Xist/XIST RNA accumulation at the Xi and altered X-linked gene expression. Our findings reveal a previously undescribed connection between NF-κB signaling pathways, which affects XCI maintenance in T cells in females.

Use of long-acting injectable cabotegravir/rilpivirine in people with HIV and adherence challenges.

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Recent changes in US Department of Health and Human Services and International AIDS Society USA guidelines now endorse use of long-acting injectable cabotegravir and rilpivirine (LA-CAB/RPV) in people with HIV (PWH) who have adherence challenges, including those with viremia. We sought to summarize clinical trial and real-world study data on outcomes and implementation strategies, highlight key unanswered questions, and provide recommendations for best practices. Studies of LA-CAB/RPV in PWH with adherence challenges demonstrate excellent virologic outcomes, although the rate of virologic failure is higher than that in registrational trials conducted in PWH with stable viral suppression. However, viral suppression is attainable on alternate antiretroviral regimen, including those that employ lenacapavir, another long-acting injectable antiretroviral drug, even after virologic failure on LA-CAB/RPV. Successful implementation strategies for long-acting programs include centralized multidisciplinary clinic teams (ideally with pharmacist/pharmacy technician involvement), small incentives to promote patient retention on injections, allowing for drop-in injections, outreach after late injections, and partnerships with home nursing, street medicine, and harm reduction sites. Creating programs that can support PWH with adherence challenges, their providers, and their clinics to use LA-CAB/RPV in service of sustained viral suppression is an urgent priority, particularly for PWH with CD4+ cell count <200 cells/mm3.

Long-acting preexposure prophylaxis: early data on roll-out in the United States.

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Long-acting preexposure prophylaxis (LA-PrEP), including cabotegravir (CAB-LA) and lenacapavir, could expand biomedical prevention coverage and reduce HIV incidence. This review describes LA-PrEP rollout in the United States, early clinical innovations in delivery, as well as opportunities and challenges for future delivery. Although CAB-LA is approved in numerous countries, availability is limited outside of implementation studies. Data on CAB-LA rollout in routine care are mainly limited to the U.S at present. Early data indicate that oral PrEP far exceeds CAB-LA use and gaps exist between prescription and receipt of CAB-LA, with barriers including insurance coverage. Successful early clinic models include multidisciplinary staffing for benefits navigation, medication procurement, and injection provision, scheduling, and monitoring. Innovative models are being explored for community health worker delivery, low-barrier care for persons with psychosocial barriers, and telehealth and community-based models. Given persistent disparities in HIV diagnoses and oral PrEP use, there is a critical need for equitable implementation of CAB-LA and forthcoming products, including long-acting lenacapavir. Gaps exist between the promise of LA-PrEP and actual use in US settings. To achieve population-level impact with LA-PrEP, there is an urgent need for greatly expanded access, clinical systems prepared for delivery, and a focus on LA-PrEP equity.

Novel Dolutegravir and Lenacapavir Resistance Patterns in Human Immunodeficiency Virus Type 2 Infection: A Case Report.

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The treatment management of human immunodeficiency virus (HIV)-2 infection presents greater challenges compared to HIV-1 infection, primarily because of inherent resistance against non-nucleoside reverse transcriptase inhibitors. Integrase strand transfer inhibitors, particularly dolutegravir, have improved treatment outcomes for people with HIV-2. Lenacapavir, a novel and potent antiretroviral capsid inhibitor, offers additional therapeutic options. However, limited knowledge exists regarding HIV-2 resistance against dolutegravir and lenacapavir. We report the case of a treatment-experienced individual who did not achieve virological suppression with regimens containing dolutegravir and lenacapavir. Clinical monitoring, genotypic and phenotypic resistance assays, and in silico structural modeling were performed. Lenacapavir was added to a failing regimen of boosted darunavir, twice daily dolutegravir, and 2 nucleoside reverse transcriptase inhibitors. Initially, this addition led to a decline in the viral load and increase in CD4+ T-cell count, despite the identification of a previously unreported combination of integrase resistance mutations. However, virological suppression was not achieved and viral load, although reduced, resumed increasing. This rebound was associated with the development of an N73D capsid substitution in HIV-2, which conferred resistance against lenacapavir. Based on cell-based assays predicting hypersusceptibility to bictegravir, the regimen was adjusted to oral lenacapavir plus bictegravir/emtricitabine/tenofovir alafenamide, resulting in a resumption in viral load decline. Although lenacapavir demonstrated therapeutic potential, our case underscores the critical need to combine it with other fully active antiretroviral agents to prevent the rapid emergence of resistance and achieve long-term virological control in treatment-experienced individuals with HIV-2.